Aakash
Medical | IIT-JEE | Foundations
Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 **NCERT Booster Test Series for NEET-2026_RM(P3)_NBTS-03C** **Time : 180 Min.**

Topics Covered:
Physics: Moving Charges and Magnetism, Magnetism and Matter, Electromagnetic Induction, Alternating Current, Electromagnetic Waves, Ray Optics & Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor, Experimental Skills
Chemistry: Alcohols, Phenols and Ethers, Aldehydes, Ketones and Carboxylic Acids, Amines (Organic Compound containing Nitrogen), The d & f-Block Elements, Coordination Compounds, The p-Block Elements, Biomolecules, Principles Related to Practical Chemistry
Botany: Principles of Inheritance & Variation, Molecular Basis of Inheritance, Microbes in Human Welfare, Organisms and Populations, Ecosystem, Biodiversity and Conservation
Zoology: Human Reproduction, Reproductive Health, Evolution, Human Health & Diseases, Biotechnology: Principles & Processes, Biotechnology & Its Applications

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. Which of the following relation represent the Biot-Savart's law?

(1) $d\vec{B} = \frac{\mu_0}{4\pi}\frac{Id\vec{l}\times\vec{r}}{r^3}$ (2) $d\vec{B} = \frac{\mu_0}{4\pi}\frac{Id\vec{l}\times\vec{r}}{r}$

(3) $d\vec{B} = \frac{\mu_0}{4\pi}\frac{Id\vec{l}\times\hat{r}}{r^3}$ (4) $d\vec{B} = \frac{\mu_0}{4\pi}\frac{Id\vec{l}\times\vec{r}}{r^4}$

2. Two particles x and y having equal charges are accelerated through the same potential difference. When they enter a region of uniform magnetic field, they describe circular path of radius R_1 and R_2 respectively. The ratio of mass of x to that of y is

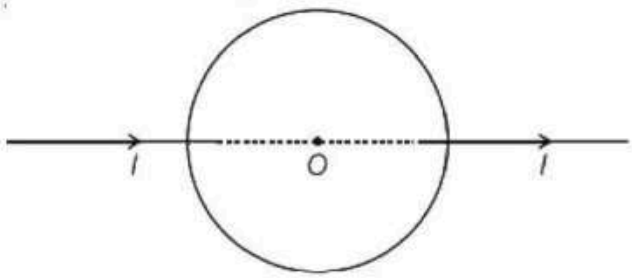
(1) $\left(\frac{R_1}{R_2}\right)^{1/2}$

(2) $\frac{R_2}{R_1}$

(3) $\left(\frac{R_1}{R_2}\right)^{2}$

(4) $\frac{R_1}{R_2}$

3. Magnetic field at the centre (O) of uniform circular coil of radius R as shown in figure is



(1) $\frac{\mu_0 I}{2R}$ (2) $\frac{\mu_0 I}{2\pi R}$

(3) Zero (4) $\frac{\mu_0 I}{4\pi R}$

4. A charged particle is released from rest in a region of steady uniform electric and magnetic fields which are parallel to each other. The particle will move in a

(1) Straight line (2) Circle

(3) Helix (4) Cycloid

5. A : The ferromagnetic substance do not obey Curie's law.
R : Above a certain temperature known as Curie point a ferromagnetic substance start behaving as a paramagnetic substance.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

6. A bar magnet having magnetic moment 10^4 J/T is free to rotate in a horizontal plane. A horizontal magnetic field $B = 4 \times 10^{-5}$ T exists in space. Work done in rotating the magnet slowly from direction parallel to field to the direction 60° from field is
(1) 0.2 J (2) 0.6 J
(3) 0.5 J (4) 0.3 J

7. The magnetic flux linked with area vector $\vec{A}$ in a uniform magnetic field $\vec{B}$ is given by
(1) A^2B (2) $\vec{A} \times \vec{B}$
(3) $\vec{A} \cdot \vec{B}$ (4) $\frac{A}{B}$

8. Two concentric circular coils one of small radius r_1 and other of large radius r_2 such that $r_1 << r_2$ are placed co-axially with their centre coinciding then mutual inductance of the arrangement is
(1) $\frac{\mu_0 \pi r_2^2}{2r_1}$ (2) $\frac{\mu_0 \pi r_1^2}{2r_2}$
(3) $\frac{\mu_0 \pi r_1^2}{r_2}$ (4) $\frac{\mu_0 \pi r_2^2}{r_1}$

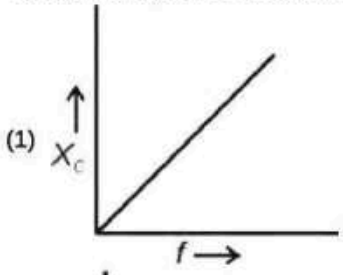
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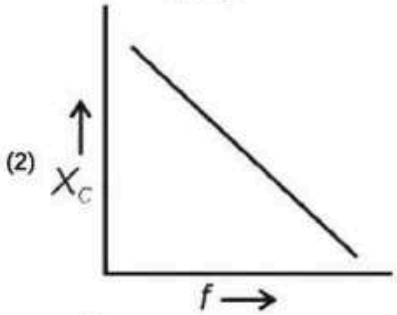
9. In a circular conducting coil, an emf of 12 V is induced when current decreases from 20 A to 2 A in 0.06 seconds. The self-inductance of the coil is
(1) 4 mH (2) 40 mH
(3) 0.8 H (4) 12 mH

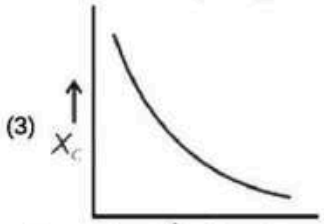
10. An alternating voltage and the current in a circuit is given by $V = 110 \sin\left(\omega t + \frac{\pi}{6}\right)$, $I = 5 \sin\left(\omega t - \frac{\pi}{6}\right)$. The average power in it is
(1) 27.5 W (2) 217.5 W
(3) 67.5 W (4) 137.5 W

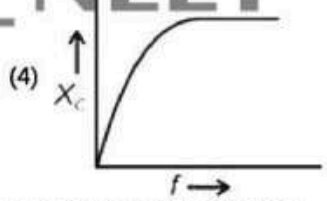
11. The incorrect statement for the state of resonance is
(1) Impedance becomes resistive
(2) The resultant reactance of the circuit is zero
(3) The phase difference between E and I is $\pi/2$ and current is maximum
(4) The power factor of the circuit becomes 1

12. Which of the following curves correctly represents the variation of capacitive reactance (X_C) with frequency f?

(1) 

(2) 

(3) 

(4) 

13. Electromagnetic waves are
(1) Deflected by electric field but not by magnetic field
(2) Deflected by magnetic field but not by electric field
(3) Deflected by both electric and magnetic field
(4) Neither deflected by electric field nor magnetic field

14. For an electromagnetic wave propagating through vacuum the amplitude of electric field vector is 12 Vm^{-1}. The amplitude of its magnetic field vector will be
(1) 4×10^{-4}T (2) 4×10^{-8}T
(3) 4×10^{8}T (4) 8×10^{-8}T

15. **Statement I :** A diminished image of an object can be obtained on a screen by a concave mirror of suitable focal length.
Statement II : The image formed by concave mirror is always real.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Statement I is correct and statement II is incorrect
(4) Statement II is correct and statement I is incorrect

16. Refraction of plane wavefront as per Huygen's principle are as shown in the figures. Figure (i) indicates refraction from air to glass and figure (ii) indicates refraction from air to water. In terms of lengths indicated in figures, the refractive index of glass w.r.t. water is

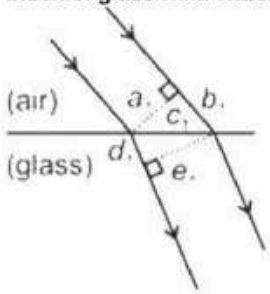


Figure (i)

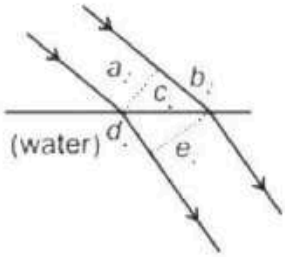


Figure (ii)

(1) $\frac{b_1}{b_2}$ (2) $\frac{d_1}{d_2}$

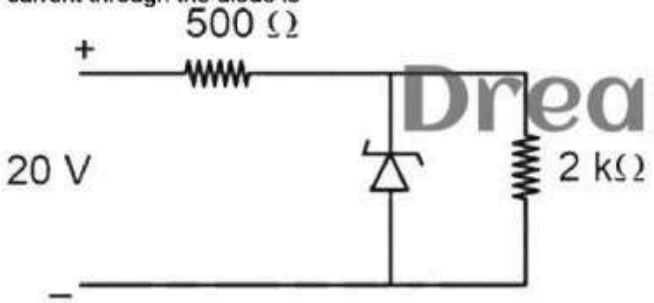
(3) $\frac{b_1 d_1}{b_2 d_2}$ (4) $\frac{b_1 d_2}{b_2 d_1}$

17. The focal length of combination of two thin lenses kept in contact is +60 cm. If the focal length of one of the lens is +25 cm, the power of the other lens will be

(1) +2.33 D (2) –2.33 D

(3) –3.73 D (4) +3.73 D

18. A Zener diode having breakdown voltage equal to 8 V, is used in a voltage regulator circuit shown in figure. The current through the diode is



(1) 10 mA (2) 15 mA

(3) 20 mA (4) 24 mA

19. Which of the following phenomenon is not common to both sound and light waves?

(1) Interference (2) Diffraction

(3) Reflection (4) Polarisation

20. In Young's double slit experiment the intensity at a point where the path difference is $\frac{\lambda}{4}$(λ being the wavelength of light used) is I. If I_0 denotes the maximum intensity, then value of $\frac{I}{I_0}$ is equal to

(1) $\frac{1}{2}$ (2) 1

(3) $\frac{1}{3}$ (4) $\frac{1}{4}$

21. The de-Broglie's wavelength associated with a particle of mass m = 2 kg and moving with a velocity 100 m/s is

(1) 6.6×10^{-34} m (2) 3.3×10^{-36} m

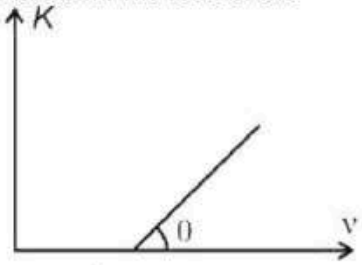
(3) 3.3×10^{-34} m (4) 5×10^{-37} m

22. If 3 eV is the energy of incident photon on metal surface and the work function of metal is 1 eV, then value of stopping potential is

(1) 1 V (2) 2 V

(3) 3 V (4) Zero

23. In photoelectric experiment, the graph of maximum kinetic energy (K) versus frequency (ν) of incident photon is plotted as shown. The value of θ is



(1) $\tan^{-1}(h)$ (2) $\tan^{-1}\left(\frac{h}{e}\right)$

(3) $\tan^{-1}\left(\frac{1}{h}\right)$ (4) $\tan^{-1}\left(\frac{e}{h}\right)$

24. de-Broglie wavelength of an electron in the first orbit of Bohr's hydrogen atom is (r = radius of orbit)

(1) πr (2) $2\pi r$

(3) $4\pi r$ (4) $\frac{\pi r}{2}$

25. The largest wavelength in Lyman series for hydrogen atom is

(1) 912 Å (2) 1615 Å

(3) 1215 Å (4) 3233 Å

26. Which of the following cannot be the angular momentum of an electron in a hydrogen atom?

(1) $\frac{2h}{2\pi}$ (2) $\frac{3h}{2\pi}$

(3) $\frac{h}{6\pi}$ (4) $\frac{h}{2\pi}$

27. The ratio of radius of nuclei $_{13}Al^{27}$ and $_{52}X^{A}$ is 3 : 5. The value of A is

(1) 81 (2) 73

(3) 65 (4) 125

28. The binding energy of deuteron $\left({}_1^2\mathrm{H}\right)$ is 1.112 MeV per nucleon and an α-particle $\left({}_2^4\mathrm{He}\right)$ has a binding energy of 7.047 MeV per nucleon. Then in the fusion reaction ${}_1^2\mathrm{H} + {}_1^2\mathrm{H} \longrightarrow {}_2^4\mathrm{He} + \mathrm{Q}$. The energy Q released is about

(1) 1 MeV (2) 11.9 MeV

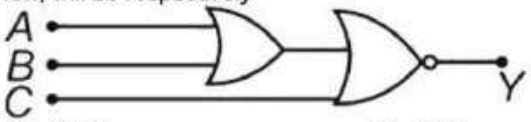
(3) 23.8 MeV (4) 931 MeV

29. The equation $_zX^A \longrightarrow {}_{z-2}Y^{A-4}$ represents

(1) α-decay (2) β-decay

(3) Fission (4) γ-decay

30. The output Y, when all three inputs are first high and then low, will be respectively



(1) (0, 1) (2) (0, 0)

(3) (1, 1) (4) (1, 0)

31. An P-type semiconductor is

(1) Positively charged

(2) Electrically neutral

(3) Negatively charged

(4) Positively or negatively charged depending upon the amount of impurity

32. Reverse bias voltage applied to a junction diode

(1) Decreases the potential barrier

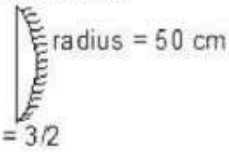
(2) Increases the potential barrier

(3) Increases the majority carrier current

(4) Decreases the minority carrier current

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33. Focal length of equivalent mirror for given silvered plano convex lens is



(1) $-\frac{25}{3}$ cm (2) $-\frac{50}{3}$ cm

(3) $\frac{25}{3}$ cm (4) $+\frac{50}{3}$ cm

34. A moving coil galvanometer has N turns and area of coil is A. If the magnetic field is B and torsional constant of suspension wire is K the current sensitivity is given by

(1) $\frac{NB}{AK}$ (2) $\frac{K}{NBA}$

(3) $\frac{NBA}{K}$ (4) $\frac{KB}{NA}$

35. If the mirror formula is given by $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$, then the error in the measurement of focal length is given by

(1) $\Delta f = f\left[\frac{\Delta u}{u^2}+\frac{\Delta v}{v^2}\right]$ (2) $\Delta f = 2f\left[\frac{\Delta u}{u^2}+\frac{\Delta v}{v^2}\right]$

(3) $\Delta f = f^2\left[\frac{\Delta u}{u^2}+\frac{\Delta v}{v^2}\right]$ (4) $\Delta f = f^2\left[\frac{\Delta u}{u^2}-\frac{\Delta v}{v^2}\right]$

36. The 'hysteresis curve' is made for

(1) Diamagnetic substance
(2) Paramagnetic substance
(3) Ferromagnetic substances
(4) All of these

37. A charged particle q having mass m is projected with velocity $\vec{v}$ in a uniform magnetic field $\left(\vec{B}\right)$ such that the velocity $\left(\vec{v}\right)$ is perpendicular to magnetic field $\left(\vec{B}\right)$. Then

(1) Force on the charged particle is constant
(2) Momentum of charged particle is constant
(3) Kinetic energy of the charged particle remains constant
(4) Work done by magnetic force on charged particle is non-zero

38. Energy stored in an inductor of inductance L will be (I is current)

(1) $2LI^2$ (2) $\frac{1}{2}LI^2$

(3) $\frac{1}{4}LI^2$ (4) $\frac{1}{8}LI^2$

39. If the separation between slits in YDSE is reduced to half, then fringe width will become

(1) Half (2) Four times
(3) Double (4) One fourth times

40. **Statement-A:** The wave nature of light shows up in the phenomenon of interference, diffraction and polarisation.
Statement-B: Photoelectric effect and Compton effect which involves energy and momentum transfer, radiation behaves as if it is made up of a bunch of particles the photon.

(1) Statement-A is correct but statement-B is incorrect
(2) Statement-A and B both are correct statements
(3) Statement-B is correct but statement-A is incorrect
(4) Statements A and B both are incorrect statements

41. A : Momentum of moving photon varies inversely as the wavelength.

R : Energy of photon is $E = mc^2$

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

42. A resistor of resistance 12 Ω and a capacitor of reactance 5 Ω are connected in series to an AC source. Impedance of the circuit will be

(1) 13 Ω (2) 7 Ω
(3) 12 Ω (4) Zero

43. Which of the following has highest penetrating power?

(1) UV-ray (2) Infrared ray
(3) Microwave ray (4) γ-ray

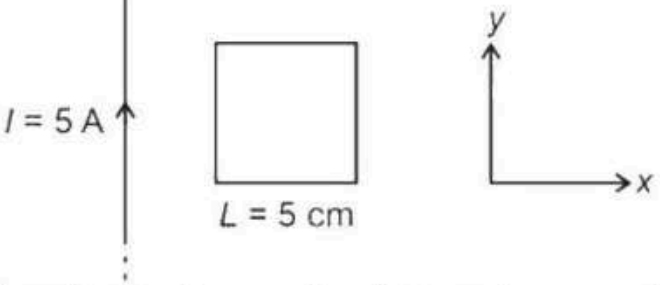
44. Which formula is used for understanding image formation in lenses?

(1) $1/f = 1/v + 1/u$ (2) $1/f = 1/v - 1/u$
(3) $f = v/u$ (4) $f = u/v$

45. This question contains 1 Match the column type question, which has 2 Columns (Column I and Column II). Column I has four entries (A), (B), (C) and (D), Column II has four entries (P), (Q), (R) and (S). Match the entries in Column I with the entries in Column II. Each entry in Column I may match with one or more entries in Column II. Four choices (1), (2), (3) and (4) are given for this question, out of which **ONLY ONE** is correct.

A current carrying long thin wire having current 5 A flowing through it, is placed near a square loop of conducting wire (side 5 cm) in xy plane as shown in the figure.

Match the changes performed given in Column-I with the corresponding results given in Column-II and choose the correct option.



	Column-I		Column-II
(A)	If the wire is moved away from the square loop	(P)	Induced current in the loop will be anticlockwise
(B)	If the magnitude of current in wire is increased	(Q)	Induced current in the loop will be clockwise
(C)	If the wire is moved towards the square loop	(R)	Wire will attract the loop
(D)	If the magnitude of current in wire is decreased	(S)	Wire will repel the loop

(1) A(P, R); B(P, S); C(Q, R); D(Q, R)
(2) A(P, S); B(Q, R); C(P, S); D(P, R)
(3) A(Q, R); B(Q, R); C(P, S); D(P, S)
(4) A(Q, R); B(P, S); C(P, S); D(Q, R)

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46. The correct order of bond dissociation enthalpy of halogens is

(1) $F_2 > Cl_2 > Br_2 > I_2$ (2) $Cl_2 > Br_2 > F_2 > I_2$

(3) $F_2 > Br_2 > I_2 > Cl_2$ (4) $F_2 > Cl_2 > I_2 > Br_2$

47. Given below are the two statements

Statement I: In P_4 molecule the bond angles are only 60°.

Statement II: White phosphorus is non-poisonous and insoluble in carbon disulphide.

In the light of the above statements, choose the correct answer.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is incorrect but statement II is correct

(4) Statement I is correct but statement II is incorrect

48. Which among the following product(s) is/are formed upon partial hydrolysis of XeF_6?

(1) $XeOF_4$ only

(2) Both XeO_3 and $XeOF_4$

(3) XeO_2F_2 only

(4) Both $XeOF_4$ and XeO_2F_2

49. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In vapour state, S_2 molecule exhibits paramagnetism.

Reason (R) : S_2 molecule has 1 unpaired electron in the antibonding π^* orbitals.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

50. The hybridisation of atomic orbitals of Xe in XeO_3, $XeOF_4$ and XeO_2F_2 are

(1) sp^3, sp^3d^2, sp^3 respectively

(2) sp^3, sp^3d^2, sp^3d respectively

(3) sp^3d, sp^3d^2, sp^3 respectively

(4) sp^3, sp^3d, sp^3d^2 respectively

51. Consider the following two statements.

Statement I: On treating a salt with conc. H_2SO_4 , evolution of brown fumes indicates the presence of nitrate ion.

Statement II: The colour of nitroso ferrous sulphate is brown.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Both statement I and statement II are correct

52. Match List-I with List-II.

List-I (Ions)	List-II (Spin only magnetic moment (BM))
a. Ti^{3+}	(i) $\sqrt{24}$
b. Mn^{3+}	(ii) $\sqrt{15}$
c. Fe^{3+}	(iii) $\sqrt{3}$
d. Co^{2+}	(iv) $\sqrt{35}$

The correct match is

(1) a(iii), b(i), c(iv), d(ii) (2) a(iii), b(iv), c(ii), d(i)

(3) a(ii), b(iii), c(i), d(iv) (4) a(ii), b(i), c(iii), d(iv)

53. Which among the following 3*d* elements has the highest enthalpy of atomisation?

(1) Ti (2) Cr

(3) V (4) Sc

54. The correct order of oxidising power of following species is

(1) $VO_2^+ > Cr_2O_7^{2-} > MnO_4^-$

(2) $MnO_4^- > VO_2^+ > Cr_2O_7^{2-}$

(3) $VO_2^+ > MnO_4^- > Cr_2O_7^{2-}$

(4) $MnO_4^- > Cr_2O_7^{2-} > VO_2^+$

55. Consider the following statements

(a) Ce(IV) is a good analytical reagent

(b) Mischmetal consists of a lanthanoid metal (~95%) and iron (~5%)

(c) Ce^{4+} and Yb^{2+} are diamagnetic species

(d) $Ce(OH)_3$ is more basic than $Tm(OH)_3$

(e) UV-visible absorption bands of lanthanoids are narrow, probably because of the excitation within f level

The correct statements are

(1) (a), (b), (c), (d) and (e) (2) (a), (b) and (d) only

(3) (c), (d) and (e) only (4) (b), (c) and (d) only

56. Given below are two statements:

Statement I: Zr and Hf have almost identical radii due to lanthanoid contraction.

Statement II: From La^{3+} to Lu^{3+} ion, ionic size decreases due to lanthanoid contraction.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

57. The correct electronic configuration of Curium (atomic number = 96) is

(1) $[Rn]5f^9 6s^1$ (2) $[Rn]5f^6 6d^2 7s^2$

(3) $[Rn]5f^7 6d^1 7s^2$ (4) $[Rn]5f^8 6s^2$

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58. Match the following aspects with their respective element.

	Aspects	Element
a.	Actinoid which reveals a maximum number of oxidation states	(i) Lutetium
b.	Lanthanoid which exhibits +4 oxidation state only in oxides	(ii) Neptunium
c.	Lanthanoid which is colourless in +3 oxidation state	(iii) Thorium
d.	Actinoid which show only one oxidation state	(iv) Terbium

Select the correct option.
(1) a(ii), b(iv), c(i), d(iii) (2) a(ii), b(i), c(iv), d(iii)
(3) a(iii), b(iv), c(i), d(ii) (4) a(iii), b(i), c(iv), d(ii)

59. Following complexes are given
$[Fe(H_2O)_6]^{3+}$ $[Fe(NH_3)_6]^{3+}$ $[Fe(CN)_6]^{3-}$
(i) (ii) (iii)
Order of stability of these complexes will be
(1) (ii) < (iii) < (i) (2) (ii) < (i) < (iii)
(3) (i) < (ii) < (iii) (4) (iii) < (ii) < (i)

60. Incorrect statement about metal carbonyl is
(1) CO acts as Lewis acid as well as Lewis base
(2) Metal acts as Lewis acid as well as Lewis base
(3) The M-C π-bond is formed by the donation of a pair of electrons from a filled d orbital of metal into the vacant antibonding π* orbital of CO
(4) The number of σ M-C bond in $[Co_2(CO)_8]$ is greater than the $[Mn_2(CO)_{10}]$

61. Correct order of C–O bond strength for the given species is
$[Cr(CO)_6]$ $[V(CO)_6]^-$ $[Mn(CO)_6]^+$
(i) (ii) (iii)
(1) (ii) > (iii) > (i) (2) (ii) > (i) > (iii)
(3) (i) > (ii) > (iii) (4) (iii) > (i) > (ii)

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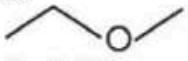
62. Given below are two statements.
Statement I: Geometrical isomerism is not possible in tetrahedral complexes.
Statement II: $[MA_3B_3]$ type complexes show geometrical isomerism.
In the light of above statements choose the correct option given below.
(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Statement I is correct but statement II is incorrect

63. Choose the incorrect match.

	Coordination compound	IUPAC names
(1)	$[Co(NH_3)_4(H_2O)Cl]Cl_2$	Tetraammine aquachloridocobalt (III) chloride
(2)	$[CoCl_2(en)_2]Cl$	Dichloridodi(ethane-1, 2-diamine) cobalt (II) chloride
(3)	$[Co(NH_3)_5(CO_3)]Cl$	Pentaamminecarbonatocobalt(III) chloride
(4)	$[Co(NH_3)_5Cl]Cl_2$	Pentaammine chloridocobalt (III) chloride

(1) (1) (2) (2)
(3) (3) (4) (4)

64. **Assertion:** Crystal field splitting in ferricyanide ion is greater than that in ferrocyanide ion.
Reason: Oxidation state of iron in ferrocyanide ion is +2 and in ferricyanide is +3.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

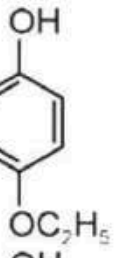
65. The ether which is most difficult to cleave on heating with HBr is
(1)  (2) $Ph–CH_2–O–CH_2–Ph$
(3) $Ph–O–CH_3$ (4) Ph – O – Ph

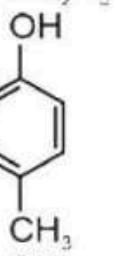
66. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R)
Assertion (A) : Ethanol is soluble in water.
Reason (R) : Ethanol forms intermolecular hydrogen bonding with water.
In the light of above statements, choose the correct option.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

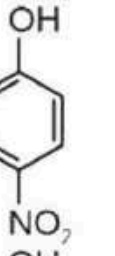
67. Which among the following reaction indicate incorrect product?
(1) $CH_3CH=CH_2 \xrightarrow{H_2O^-} CH_3\underset{OH}{\underset{|}{C}}H-CH_3$
(2) $CH_3CH=CH_2 \xrightarrow[(ii)\ H_2O_2/OH^-]{(i)\ B_2H_6/THF} CH_3\underset{OH}{\underset{|}{C}}H-CH_3$
(3) $CH_2CH_2CHO \xrightarrow{H_2/Pd} CH_2CH_2CH_2OH$
(4) $CH_3CH_2COCH_2CH_3 \xrightarrow{NaBH_4} CH_3CH_2\underset{OH}{\underset{|}{C}}HCH_2CH_3$

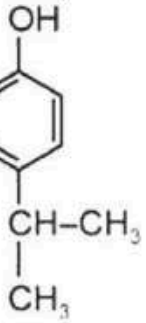
68. $\text{Propionicacid} \xrightarrow{PCl_5} X \xrightarrow[\text{Anhyd. } AlCl_3]{C_6H_6} Y.$
In the above sequence Y is
(1) Phenylethylketone (2) Phenylacetate
(3) Ethyl benzene (4) Ethyl benzoate

69. Which of the following substituted phenols is the strongest acid?

(1) 

(2) 

(3) 

(4) 

70. Consider the following reactions.

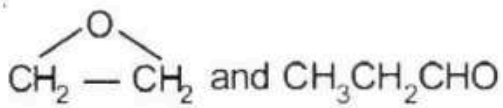
(i) $CH_3CH_2 - OH \xrightarrow{PCC}$ product (A)]

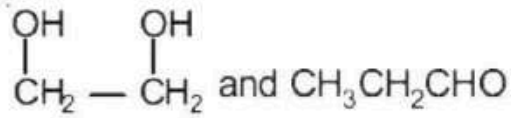
(ii) $CH_3CH_2CH_2 - OH \xrightarrow[573\ K]{Cu}$ product (B)

Products A and B respectively are

(1) CH_3CHO and CH_3CH_2CHO

(2) CH_3CHO and CH_3COCH_3

(3) 

(4) 

71. Correct order of acidic strength of the given compounds is

(i) FCH_2COOH

(ii) NO_2CH_2COOH

(iii) $NC-CH_2COOH$

(1) (ii) > (i) > (iii)
(2) (i) > (ii) > (iii)
(3) (ii) > (iii) > (i)
(4) (iii) > (i) > (ii)

72. Consider the following statements

a. Acetaldehyde gives positive 2,4-DNP test

b. Tollens' reagent oxidises aldehyde to corresponding carboxylate anion

c. Benzoic acid does not undergo Friedel-Crafts reaction

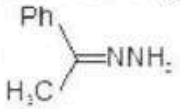
The correct statements are

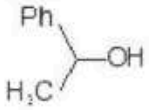
(1) a and b only
(2) b and c only
(3) a and c only
(4) a, b and c

73. When acetophenone is treated with hydrazine followed by heating with sodium hydroxide in presence of ethylene glycol then the end product formed is

(1) $PhCH_2CH_3$

(2) $Ph - CH = CH_2$

(3) 

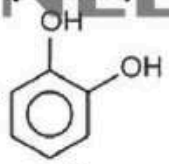
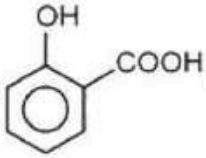
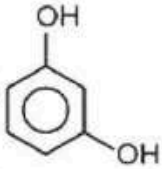
(4) 

74. Which of the following organic compound(s) undergo(es) disproportionation reaction with conc. NaOH?

(A) Propionaldehyde
(B) Formaldehyde
(C) Butanone
(D) Benzaldehyde

(1) (A) only
(2) (B) and (D) only
(3) (A), (B) and (C) only
(4) (A), (C) and (D) only

75. Match List-I with List-II.

List I (Structure)	**List-II (Common name)**
a. 	(i) Resorcinol
b. 	(ii) Quinol
c. 	(iii) Catechol
d. 	(iv) Salicylic acid

Choose the correct answer from the options given below

(1) a(iv), b(iii), c(i), d(ii)
(2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(iv), c(ii), d(i)
(4) a(iii), b(iv), c(i), d(ii)

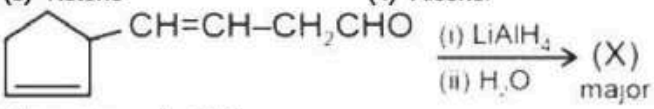
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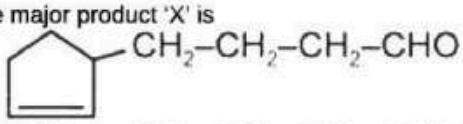
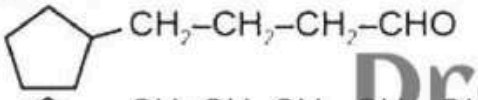
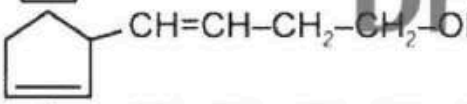
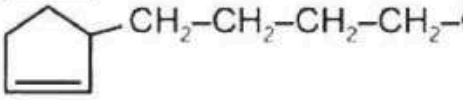
76. Given below are two statements, one is labelled as assertion (A) and the other is labelled as reason (R).
Assertion (A): Boiling point of carboxylic acids are higher than aldehydes and ketones of comparable molecular mass.
Reason (R) : Carboxylic acids have extensive association through intermolecular hydrogen bonding.
In the light of above statements, select the most appropriate option.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

77. Consider the following reaction sequence

$$\text{Propan-1-ol} \xrightarrow{SOCl_2} \underset{\text{(Major)}}{A} \xrightarrow{KCN} \underset{\text{(Major)}}{B} \xrightarrow[(ii)\ H_3O^+]{(i)\ CH_3MgBr} \underset{\text{(Major)}}{C}$$

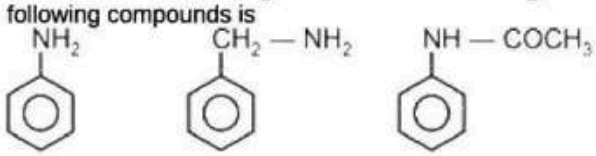
Major product C is a/an
(1) Ether (2) Aldehyde
(3) Ketone (4) Alcohol

78. 

The major product 'X' is
(1) 
(2) 
(3) 
(4) 

79. Benzylamine and aniline can be chemically distinguished by
(1) Hinsberg's reagent (2) Br_2/water
(3) Dilute H_2SO_4 (4) $CHCl_3$/KOH/Δ

80. The correct decreasing order of basic strength for the following compounds is



(I) (II) (III)
(1) I > II > III (2) III > I > II
(3) II > I > III (4) II > III > I

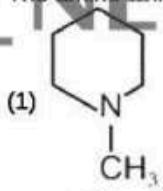
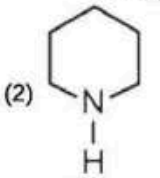
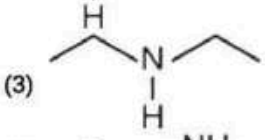
81. Given below are the two statements. One is labelled as assertion (A) and other is labelled as reason (R).
Assertion (A): Ammonolysis of excess of alkyl halide at 373 K produces a mixture of 1° amine, 2° amine, 3° amine and quaternary ammonium salt.
Reason (R): NH_3 firstly reacts with RX to produce 1° amine. The 1° amine thus obtained behaves as a nucleophile and can further react with alkyl halide to form 2° and 3° amines, and finally quaternary ammonium salt.
In the light of above statements, choose the correct option.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) (A) is true but (R) is false
(3) (A) is false but (R) is true
(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

82. Find the incorrect match.

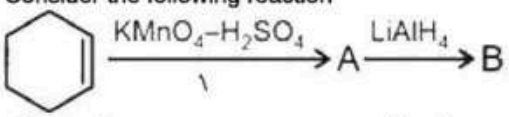
	Column I	Column II
(1)	Keratin	– Fibre-like structure
(2)	Insulin	– Spherical shape
(3)	Albumin	– Insoluble in water
(4)	Myosin	– Hydrogen and disulphide bonds

(1) (1) (2) (2)
(3) (3) (4) (4)

83. Which of the following reagents does not confirm the presence of a carbonyl group $\left(\begin{matrix} \\ H \end{matrix}\!\!>\!C=O\right)$ in glucose?
(1) NH_2OH (2) Br_2, water
(3) Acetic anhydride (4) HCN

84. The amine which will not react with Hinsberg's reagent is
(1) 
(2) 
(3) 
(4) 

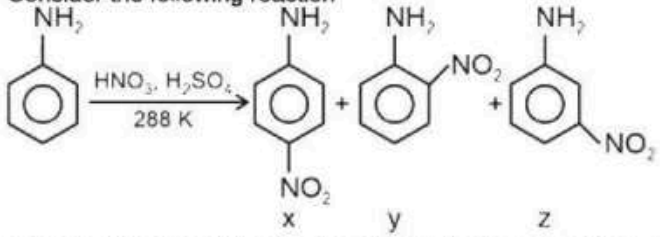
85. Consider the following reaction



Choose the correct statement regarding the product 'B'
(1) Product 'B' gives orange-red ppt with 2,4-DNP reagent
(2) Product 'B' is adipic acid
(3) Product 'B' gives H_2 gas upon reaction with Na
(4) Product 'B' gives iodoform test

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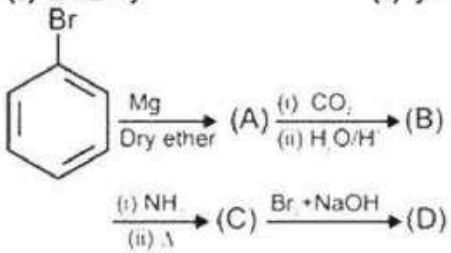
86. Consider the following reaction



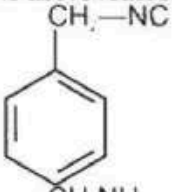
Correct order regarding the % yield of the formed products will be

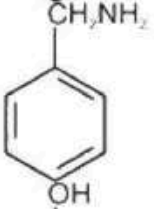
(1) x > y > z (2) z > y > x
(3) x > z > y (4) y > z > x

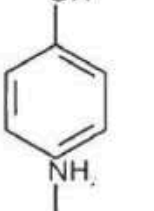
87.

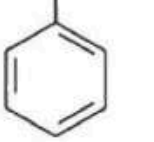


In the above reaction sequence, D is

(1) 

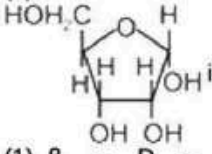
(2) 

(3) 

(4) 

88. Water-soluble and fat-soluble vitamins respectively are

(1) Vitamin – C and Vitamin – B_5
(2) Vitamin – D and Vitamin – E
(3) Vitamin – B_2 and Vitamin – A
(4) Vitamin – K and Vitamin – C

89.  is known as

(1) β – D – 2 – deoxyribose (2) α – D – 2 – deoxyribose
(3) β – D – ribose (4) α – D – ribose

90. Number of chiral carbons present in α-D-glucose is

(1) 4 (2) 5
(3) 6 (4) 3

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BOTANY

91. An age pyramid of a population is said to be declining if

(1) The reproductive population is approximately equal to pre-reproductive population
(2) There is fewer individuals in pre-reproductive group as compared to reproductive group
(3) The post-reproductive population is less than the reproductive population
(4) There is more individuals in pre-reproductive group as compared to reproductive group

92. Identify the following statements as true **(T)** or false **(F)** and select the **correct** option.
A. Failure of segregation of homologous chromosomes during cell division leads to aneuploidy.
B. Polyploidy is more common in plants.
C. Non-haemophilic parents cannot have a haemophilic child.

	A	B	C
(1)	F	T	T
(2)	T	F	T
(3)	T	T	F
(4)	T	F	F

(1) (1) (2) (2)
(3) (3) (4) (4)

93. Law of independent assortment can be explained with the help of
(1) Test cross (2) Back cross
(3) Dihybrid cross (4) Monohybrid cross

94. Read the following statements and choose the **correct** option.
Statement-A: Organic farmers work to create a system which will keep pest at manageable level by complex system of checks and balances within vibrant ecosystem.
Statement-B: Biocontrol refers to the use of biofertilizer that will enrich the nutrient quality of the soil.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

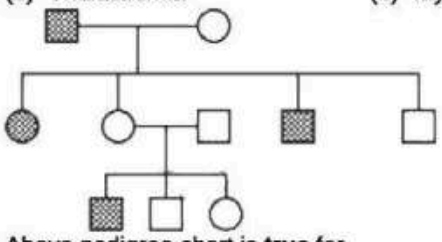
95. Karyotype of an individual who is inflicted with Turner's syndrome is
(1) 45 + X0 (2) 45 + XX
(3) 44 + X0 (4) 45 + XXY

96. Two tall pea plants are crossed together. In its F_1 generation, the probability of tallness will be
(1) 100% tall or 75% tall (2) Always 100% tall
(3) 100% tall or 50% tall (4) Always 50% tall

97. The best example for pleiotropy is
(1) ABO Blood group (2) Skin colour
(3) Phenylketoneuria (4) Colour Blindness

98. Choose **odd** one w.r.t. recessive Mendelian disorders.
(1) Haemophilia (2) Colour blindness
(3) Thalassemia (4) Myotonic dystrophy

99.



Above pedigree chart is **true** for
a. X-linked recessive inheritance
b. Autosomal recessive inheritance
c. Autosomal dominant inheritance
(1) Only a (2) a and b
(3) b and c (4) Only b

100. Read the statement (A-D) and select the **correct** ones w.r.t. morgan experiments on linkage
(A) When genes were grouped on the same chromosome, some genes were loosely linked and showed very low recombination.
(B) The genes for white eye colour and yellow body colour were tightly linked and showed 1.3 percent recombination.
(C) The genes for white eye colour and miniature wings were loosely linked and showed 37.2% parental types.
(D) Morgan found that proportion of parental gene combination was much higher than non-parental gene combination.
(1) Statements (A) & (C) (2) Statements (B), (C) & (D)
(3) Statement (B) only (4) Statements (B) & (D)

101. R strain of *Streptococcus pneumoniae*
(1) Produces smooth colonies on a culture plate
(2) Has mucous or polysaccharide coat
(3) Does not cause pneumonia in mice
(4) Is the virulent strain of bacteria

102. Which of the following is **not** a part of transcription unit in DNA?
(1) Promoter (2) Terminator
(3) Repressor (4) Structural gene

103. Select the **incorrect** statement regarding the salient features of the double-helix structure of DNA
(1) It is made of two polynucleotide chains
(2) The two chains have anti-parallel polarity
(3) Guanine is bonded with thymine by three H-bonds
(4) The two chains are coiled in a right-handed fashion

104. Select the **incorrect** statement from the following.
(1) In most of the eukaryotes, the structural gene in monocistronic
(2) Exons are the expressed sequence of genes
(3) Introns do not appear in processed RNA
(4) A gene does not code for tRNA

105. In eukaryotes, RNA polymerase II synthesises
(1) 28 S rRNA (2) hn rRNA
(3) t-RNA (4) snRNA

106. A molecule that can act as a genetic material should
(1) Provide scope for rapid changes
(2) Be unable to generate its replica
(3) Function as a catalyst
(4) Not change its nature with the different stages of life cycle

107. Select the **correct** statements for gross primary productivity (GPP) of an ecosystem.
a. Rate of formation of new organic matter by consumers.
b. A considerable amount of GPP is utilised by plants in respiration.
c. Rate of production of organic matter during photosynthesis per unit time and area.
d. GPP is always more than NPP (Net primary productivity).
(1) a & b only (2) b & c only
(3) All except a (4) All except a & c

108. A genetic material contains guanine (G) = 14%, which of the following base proportion confirms that it is double stranded DNA?
(1) T = 36%, C = 14%, A = 36%
(2) C = 36%, T = 14%, A = 14%
(3) A = 14%, T = 36%, C = 36%
(4) U = 36%, C = 14%, A = 36%

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109. State **true (T)** or **false (F)** for the given statements and choose the option accordingly.
a. In bacteria, translation can begin much before mRNA is fully formed.
b. In the process of tailing of hnRNA, adenylate residues (250-300) are added at the 3' in a template dependent manner.
c. The genetic code is absolutely universal.
d. The presence of introns in split-gene is reminiscent of antiquity.
(1) a(F), b(T), c(T), d(T) (2) a(T), b(F), c(F), d(T)
(3) a(T), b(F), c(T), d(F) (4) a(F), b(T), c(F), d(F)

110. Read the following statements and select the **correct** option.
Assertion (A): Propagation of piece of DNA during recombinant DNA procedures, requires a vector.
Reason (R): Vector provides the origin of replication.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

111. Match the following columns and select the **correct** option.

	Column		**Column II**
a.	Har Gobind Khorana	(i)	His cell-free system for protein synthesis helped to decipher the genetic code
b.	George Gamow	(ii)	Developed chemical method that was instrumental in synthesising RNA molecules with defined combinations of bases
c.	Marshall Nirenberg	(iii)	Polynucleotide phosphorylase helpful in polymerising RNA with defined sequence in a template independent manner
d.	Severo Ochoa	(iv)	A physicist who suggested that in order to code for all the 20 amino acids, the code should be made up of three nucleotides.

(1) a(i), b(iii), c(ii), d(iv) (2) a(ii), b(i), c(iv), d(iii)
(3) a(iii), b(ii), c(iv), d(i) (4) a(ii), b(iv), c(i), d(iii)

112. Floating debris from sewage is removed by
(1) Sequential filtration (2) Sedimentation
(3) Sludge digesters (4) Aeration tank

113. Aeration tank of sewage treatment plant
(1) Is used for removal of soil and small pebbles
(2) Has activity of methanogens for production of biogas
(3) Allows growth of useful aerobic bacteria associated with fungal filaments
(4) Shows microbial activity that leads to rise in BOD of water

114. Match the following columns and select the **correct** option.

	Column-I		**Column-II**
a.	*Trichoderma*	(i)	Controls butterfly caterpillars
b.	*Bacillus thuringiensis*	(ii)	Effective against several plant pathogens
c.	Ladybird beetle	(iii)	Pathogen that attacks specific insect
d.	Baculovirus	(iv)	Controls aphids

(1) a(iii), b(iv), c(i), d(ii) (2) a(ii), b(i), c(iii), d(iv)
(3) a(iii), b(i), c(iv), d(ii) (4) a(ii), b(i), c(iv), d(iii)

115. Puffed-up appearance of dough during fermentation is due to production of
(1) Ethyl alcohol (2) Lactic acid
(3) Water vapour (4) Carbon dioxide

116. Read the following statements and select the **correct** option regarding lactic acid bacteria (LAB).
Statement-A : LAB produce acids that coagulate and completely digest the milk proteins.
Statement-B : LAB play very beneficial role in checking disease-causing microbes in our stomach.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are incorrect
(4) Both statements A and B are correct

117. Read the following statements :
(a) It is produced by a bacterium.
(b) It is used as an immunosuppressive agent.
(c) It is useful for patients who have undergone myocardial infarction.
(d) It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.
Which amongst the given statements is/are **true** for streptokinase?
(1) (a) and (d) (2) (a) and (c)
(3) (b) and (d) (4) (c) only

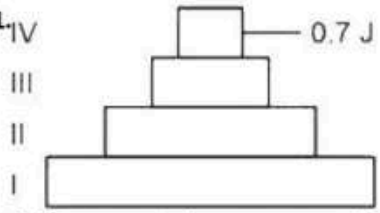
118. Which of the following statement is **incorrect**?
(1) 'Toddy' a traditional drink is made by fermenting sap from palms
(2) The Roquefort cheese are ripened by growing a specific fungi on them
(3) Alexander Fleming discovered penicillin while working on *Streptococci*
(4) *Bacillus thuringiensis* are available in sachets as dried spores which are mixed with water and sprayed onto vulnerable plants

119. Ecologically, standing crop means the amount of living material present in
(1) Trophic level of green plants only at a given time
(2) Same trophic levels of only consumers per unit area
(3) Different trophic levels at a given time per unit area
(4) Same trophic level of only primary consumers per unit area

120. The interaction is detrimental for both of the interacting species in
(1) Competition (2) Parasitism
(3) Mutualism (4) Commensalism

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121.



If the amount of energy present at trophic level IV is 0.7 J then the amount of energy present at trophic level III in the above pyramid of energy is

(1) 7 J (2) 700 J
(3) 0.7 J (4) 7000 J

122. If 6 individuals in a laboratory population of 24 *Drosophila* died in one week. What will be the mortality rate per *Drosophila* per week?

(1) 0.4 (2) 0.25
(3) 0.1 (4) 0.18

123. Which of the given relationships or associations may show co-extinction if one partner is eliminated in nature?

(1) Fig and fig wasp
(2) Nile perch and cichlid fish
(3) Water hyacinth and aquatic organism
(4) African catfish and Indian catfish

124. Select **incorrect** statement w.r.t. Mac Arthur experiment on five closely related species of warblers.

(1) They were living together on the same tree
(2) They were able to avoid competition
(3) They were unable to promote co-existence
(4) They enhanced their rate of survival by changing foraging patterns

125. Read the following statements and select the **correct** option.

a. Monarch butterfly acquires toxic chemicals during the caterpillar stage.
b. Competition cannot occur between two unrelated species.
c. Barnacle *Balanus* dominated over the *Chathamalus* in the intertidal area of the rocky sea coasts of Scotland.
d. Darwin was convinced that interspecific competition is a potent force in organic evolution.

(1) All are true except (b) (2) Only (a) and (c) are false
(3) (b), (c) and (d) are true (4) Only (a) and (b) are false

126. Read the following statements and select the **correct** option

Statement A: Colombia located near the equator has nearly 1,400 species of birds while New York at 41° N has 105 species.

Statement B: India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds.

(1) Both the statements A and B are incorrect
(2) Only statement A is correct
(3) Only statement B is correct
(4) Both the statements A and B are correct

127. Choose the **correct** set w.r.t. criterias of "Biodiversity hotspot"

(a) High level of species richness.
(b) High degree of endemism.
(c) Regions of accelerated habitat loss.

(1) Only (a) and (b) (2) Only (b) and (c)
(3) Only (a) and (c) (4) All (a), (b) and (c)

128. Select the **odd** one for *in-situ* conservation of biodiversity?

(1) Biosphere reserves (2) Wildlife safari parks
(3) National parks (4) Wildlife sanctuaries

129. Most important cause for biodiversity loss is

(1) Over exploitation
(2) Alien species invasion
(3) Co-extinctions
(4) Habitat loss and fragmentation

130. Consider the following statements and state them as True **(T)** or False **(F)** and choose the **correct** option.

A. At ecosystem level, India has greater ecosystem diversity than the Scandinavian countries.
B. Species diversity increases as we move from the equator towards the pole
C. Extinction of Steller's sea cow and passenger pigeon were due to their over exploitation by humans.
D. India has only 8.1 per cent of the world's land area, its share of the global species diversity is an impressive 2.4 per cent.

	A	B	C	D
(1)	T	T	F	F
(2)	F	F	F	T
(3)	T	T	T	F
(4)	T	F	T	F

(1) (1) (2) (2)
(3) (3) (4) (4)

131. A predator

(1) Is too efficient to overexploit its prey
(2) Helps in maintaining species diversity by increasing the intensity of competition among prey species.
(3) Acts as conduit for energy transfer across trophic levels
(4) Shows (+, +) interaction with its prey

132. Read the statements given below and choose the **correct** option.

Statement I: Decomposition is purely an anaerobic process.
Statement II: The rate of decomposition is controlled by chemical composition of detritus and climatic factors.

(1) Only statement I is correct
(2) Only statement I is incorrect
(3) Both statements I and II are correct
(4) Both statements I and II are incorrect

133. Which of the following ecological pyramids is **not** possible for a tree ecosystem?

(1) Inverted pyramid of number
(2) Upright pyramid of energy
(3) Spindle-shaped pyramid of number
(4) Inverted pyramid of biomass

134. Mark the **incorrect** statement w.r.t. ecological pyramid.

(1) It assumes a simple food chain
(2) Decomposers are not given any place
(3) It does not take into account the same species belonging to two or more trophic levels
(4) The base represents the first level of consumers

135. Energy captured by plants to perform photosynthesis is approximately

(1) 2-10% of incident solar radiation
(2) 2-10% of photosynthetically active radiation
(3) 50% of incident solar radiation
(4) 50% of photosynthetically active radiation

ZOOLOGY

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136. Extrusion of second polar body from egg nucleus occurs
(1) After entry of sperm and before completion of fertilization
(2) After completion of fertilization
(3) Before the entry of sperm
(4) Without any relation with sperm entry

137. In a hypothetical condition, if a 5-weeks pregnant female consumes a drug which inhibits the secretion of hCG, then
(1) Pregnancy will still continue normally
(2) Corpus luteum will still continue secreting progesterone at normal levels
(3) Placenta will start secreting high levels of progesterone immediately for compensating the loss of hCG
(4) Endometrium will start shedding off due to decrease in progesterone levels

138. After ovulation, the ruptured follicle is converted into a structure called corpus luteum, which mainly secretes the hormone_____.
Choose the option that fills the blank **correctly**.
(1) Testosterone (2) Prolactin
(3) Progesterone (4) Oxytocin

139. During cleavage in embryonic development, there is increase in all of the following, **except**
(1) Number of blastomeres (2) Size of blastomeres
(3) Nuclear cytoplasmic ratio (4) DNA content

140. Choose the **incorrect** match w.r.t. human female reproductive system.
(1) Mons pubis – A cushion of fatty tissue covered with skin and pubic hair
(2) Labia majora – Fleshy folds of tissue extending down from mons pubis
(3) Ampulla of oviduct – Site of fertilization
(4) Mammary lobes – 40-50 in each breast

141. Which of the following is **incorrect** difference between spermatogenesis and oogenesis?

	Spermatogenesis	Oogenesis
(1)	Occurs in testis	Occurs in ovary
(2)	Spermatogonia form sperms	Oogonia form ova
(3)	Both maturation divisions are completed in testis	Both maturation divisions are completed in ovary
(4)	Sperms reduce in size from spermatogonia	Ova increases in size from oogonia

(1) (1) (2) (2)
(3) (3) (4) (4)

142. Read following statements carefully and choose the option which represents each statement as true (T) or false (F) **correctly**
A. Chorionic villi appear on trophoblast before implantation
B. Morula is transformed into blastocyst as it moves across the fallopian tube
C. Blastocyst becomes embedded in myometrium of uterus due to rapid division of uterine cells
D. The placenta is connected to the embryo through an umbilical cord

	A	B	C	D
(1)	T	F	T	F
(2)	F	T	F	T
(3)	T	T	T	F
(4)	F	F	F	T

(1) (1) (2) (2)
(3) (3) (4) (4)

143. Read the following statements.
a. First meiotic division of primary oocyte is completed in tertiary follicle inside ovary.
b. Mesometrium layer of uterus undergoes strong contractions during parturition.
c. During cleavage divisions in a human embryo, there is a decrease in DNA : cytoplasm ratio.
d. In all mammals, testes are present in scrotal sac outside their body during breeding season.
How many among the above mentioned statements is/are **incorrect**?
(1) One (2) Two
(3) Three (4) Four

144. Consider the following statements.
(a) Barrier methods prevent physical meeting of gametes but don't affect gamete formation.
(b) CuT increases the phagocytosis of sperms in uterus and suppresses motility and fertilising capacity of sperms.
(c) Saheli does not provide protection against STDs.
(d) MTP is relatively safe during first trimester of pregnancy.
Select the option with **correct** set of statements.
(1) (a) and (b) only (2) (b) and (d) only
(3) (a), (c) and (d) only (4) (a), (b), (c) and (d)

145. An ART, in which embryo with more than 8 blastomeres could be transferred into the uterus is
(1) ZIFT (2) IUI
(3) IUT (4) AI

146. Select the **mismatch** w.r.t. mode of action.
(1) Oral pills – Inhibit ovulation and implantation
(2) Vasectomy – Block gamete transport and gametogenesis
(3) Diaphragms – Prevent physical meeting of male and female gametes
(4) LNG 20 – Alters the quality of cervical mucus and makes cervix hostile to sperms

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147. Read the following statements.
a. Sterilisation procedure in males is called 'vasectomy'.
b. Chlamydiasis is a bacterial disease.
c. Hepatitis B is an incurable disease.
d. Family planning programmes in India were initiated in 1951.
How many of the above statements are **correct**?
(1) One (2) Three
(3) Four (4) Two

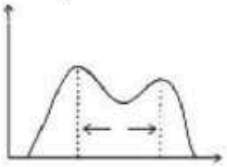
148. Factor which **did not** contribute to increase in population size of India during last few decades is
(1) Decline in death rate
(2) Increase in maternal mortality rate (MMR)
(3) Decline in infant mortality rate (IMR)
(4) Increase in number of people in reproducible group

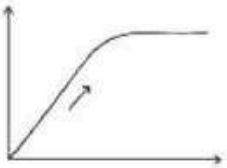
149. Select the **incorrect** statement w.r.t. contraceptive device.
(1) Combined pill does not inhibit ovulation
(2) Cu-T inhibits fertilization
(3) Morning after pill is effective within 72 hours of unprotected sex
(4) Spermicidal creams, jellies and foams are seldom used alone due to high failure rate

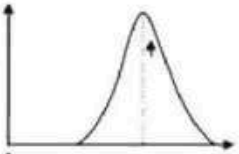
150. Which of the following is/are **incorrect** statements about the observation/work of Darwin and *Hugo de Vries*?
(1) Charles Robert Darwin clearly and convincingly set forth the concept of natural selection as the mechanism of evolution, *de Vries* gave mutation theory
(2) Darwin emphasised the adaptive nature of variation, *de Vries* stressed on its randomness
(3) According to *Hugo de Vries* variations are small and directional, but Darwinian variations are random and directionless
(4) Evolution for Darwin was gradual while *de Vries* believed mutation caused speciation and hence called it saltation

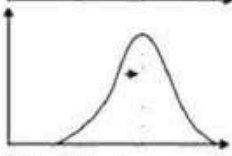
151. Read the given statements and select the **correct** option
Statement-A : Phenomenon like habitat fragmentation and genetic drift may accentuate variations leading to speciation and evolution.
Statement-B : Due to continental drift, eutherians in Australia survived and continue to do so in wild because of lack of competition from metatherians.
(1) Only statement A is correct
(2) Both statements A and B are correct
(3) Only statement B is correct
(4) Both statements A and B are incorrect

152. Industrialisation in Britain generated smoke and soot that led to the selection of melanised moths against the non-melanised moths. Which of the following types of natural selection represents the above example?

(1) 

(2) 

(3) 

(4) 

153. All of the following are the examples of the effect of use and disuse of organs as per Lamarckism, **except**
(1) Elongation of neck in giraffe
(2) Disappearance of limbs in snakes
(3) Blind cave dwellers
(4) Elongation of tail of tadpole during metamorphosis

154. According to Hardy-Weinberg equilibrium, the conditions necessary for gene frequencies to remain constant are all **except**
(1) The population must be very large
(2) Mating must be completely random
(3) Migration of individual organisms into and out of the population must not occur
(4) Mutations should be frequent

155. In 1953, S.L. Miller observed the formation of amino acids after conducting an electrical discharge experiment in a closed flask which contained
(1) Alanine, glycine, aspartic acid, water vapour
(2) O_2, CO_2, H_2O, N_2
(3) CH_4, H_2, NH_3, water vapour
(4) O_2, CO_2, NH_3, H_2O

156. Read the following statements A and B and choose the **correct** option.
Statement A: Adaptive radiation is the process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography.
Statement B: Australian marsupial (Bobcat) and placental mammal (Tasmanian tiger cat) are example of divergent evolution.
(1) Statement A is correct but statement B is incorrect
(2) Statement A is incorrect but statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

157. Select **incorrect** statement w.r.t. Charles Darwin.
(1) H.M.S. Beagle is name of sail ship used in his voyage
(2) Branching descent and natural selection are two key concepts of Darwin's theory
(3) The geological history of earth is not closely correlated with the biological history of earth
(4) The fitness, ultimately refers to only reproductive fitness

158. In AIDS patients, both cell mediated immunity and antibody mediated immunity are nearly lost due to severe decrease in number of
(1) B-lymphocytes (2) Helper T-cells
(3) NK cells (4) Macrophages

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159. How many among the given statements is/are **correct**?
a. Alcohol addicts and all types of drug abusers are at high risk for acquiring HIV infection.
b. Chemical carcinogens like UV rays and X-rays cause neoplastic transformations.
c. Cancer is the most dreaded infectious disease of human.
d. Drugs used in chemotherapy besides targeting cancer cells, also inhibit the proliferation of normal cells in body leading to hair loss and anaemia.
(1) Four (2) Three
(3) Two (4) One

160. Which of the following cancer detection methods uses strong magnetic fields and non-ionising radiations to detect pathological changes in the living tissue?
(1) CT (2) MRI
(3) X-rays (4) Chemotherapy

161. Acidic pH in stomach, saliva in mouth and tears from the eyes are parts of
(1) Physical barriers of innate immunity
(2) Physiological barriers of innate immunity
(3) Cytokine barriers of innate immunity
(4) Humoral/fluid mediated specific immunity

162. **Assertion (A):** A 70 year old man's immunity is weaker in comparison to that of a 25-year old man's immunity.
Reason (R): The size and secretion of thymus gland is affected by the age of the individual.
In the light of above statements, choose the **correct** answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the explanation of (A)
(3) (A) is true and (R) is false
(4) (A) is false and (R) is true

163. A is a depressant while B has stimulating action on CNS.
Select the correct option to fill in the blanks 'A' and 'B' respectively.
(1) Cocaine, morphine (2) Heroin, cocaine
(3) Ganja, heroin (4) Heroin, morphine

164. Which of the following is responsible for the chills and high fever recurring every 3 to 4 days in case of malarial infection?
(1) Rapid multiplication of sporozoites within hepatocytes
(2) Infection of sporozoites with the bite of mosquito to the host
(3) Rupture of RBCs that leads to the release of haemozoin
(4) Release of parasite that infects new RBCs in human blood

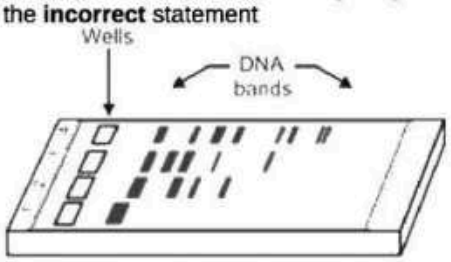
165. 'DNA cannot pass through cell membranes.' The given characteristic is due to the presence of _____ in DNA.
Select the **correct** option to fill in the blank.
(1) Hydrophilic moiety
(2) Hydrophobic moiety
(3) Large structural components
(4) Nitrogenous bases

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166. From given statements select the **incorrect** statement.
(a) Proteins which are obtained by recombinant DNA technology are called recombinant proteins
(b) Downstream processes include separation and purification of the desired product
(c) After purification, the product has to be formulated with suitable preservatives
(d) Downstream processing and quality control testing do not vary from product to product
(1) (a) (2) (b)
(3) (c) (4) (d)

167. If gene of interest is inserted at *Sal* I site in pBR322, then recombinant is resistant for
(1) Kanamycin
(2) Ampicillin and tetracycline
(3) Ampicillin
(4) Tetracycline

168. A circular plasmid was taken and subjected to restriction digestion with multiple restriction enzymes.
The resultant mixture was subjected to agarose gel electrophoresis. Observe the figure given below and choose the **incorrect** statement



(1) Possibly, completely digested sample is in lane 4
(2) Largest DNA fragment is closest to the wells
(3) Undigested DNA sample is in lane 2
(4) Smallest DNA fragment is closest to anode

169. The separated bands of DNA are cut from the agarose gel and extracted from the gel piece. This step is known as
(1) Spooling (2) Denaturation
(3) Elution (4) Fragmentation

170. Plant cells are bombarded with high velocity microparticles of X coated with Y in a method called Z. Choose the option that fill the blanks **correctly.**

	X	Y	Z
(1)	Gold or tungsten	DNA	Biolistics
(2)	Platinum and gold	RNA	Biolistics
(3)	Gold or silver	RNA	Gene gun
(4)	Platinum or silver	RNA	Gene gun

(1) (1) (2) (2)
(3) (3) (4) (4)

171. The ends of DNA fragments produced after RE digestion are sticky due to the presence of
(1) Unpaired bases (2) Calcium ions
(3) Paired Pyrimidines (4) Free methyl group

172. If the gene of interest is cloned at *Eco*RI in pBR322, the recombinant *E-coli* after transformation are
(1) Susceptible to ampicillin and tetracyline
(2) Sensitive to tetracycline
(3) Resistant to kanamycin
(4) Resistant to ampicillin and tetracycline

173. A single PCR cycle involves three temperature dependent steps. Their **correct** sequence is
(1) Annealing →denaturation →amplification
(2) Denaturation →annealing →extension
(3) Extension →annealing →denaturation
(4) Annealing →extension →denaturation

174. The construction of the first recombinant DNA emerged from the possibility of linking a gene-encoding antibiotic resistance with a native plasmid of
(1) *Bacillus thuringiensis*
(2) *Salmonella typhimurium*
(3) *Escherichia coli*
(4) *Agrobacterium tumefaciens*

175. How many of the following is/are correct w.r.t. Bt toxin?
(a) In *Bacillus thuringiensis*, it exists as protoxin
(b) Alkaline pH of dipteran's gut favors its activation
(c) The activated toxin binds to mesothelial cells of foregut
Choose the **correct** option.
(1) Zero (2) Two
(3) One (4) Three

176. **Assertion (A):** Bacteria can be made to carry genes coding 'A' and 'B' chains of insulin for the synthesis of human insulin.
Reason (R): Genetic code is nearly universal.
In the light of above statements, select the **correct** option.
(1) Both (A) and (R) are true, (R) correctly explains (A)
(2) Both (A) and (R) are true, (R) does not correctly explain (A)
(3) (A) is true, (R) is false
(4) (A) is false, (R) is true

177. The capacity to generate a whole plant from any cell/explant is called
(1) Somatic hybridisation (2) Totipotency
(3) Micropropagation (4) Polymerisation

178. Golden rice is genetically modified and is enriched in the precursor of
(1) β- galactosidase (2) Vitamin A
(3) Vitamin B_{12} (4) α-lactalbumin

179. Which among the following methods involve silencing of a specific mRNA due to formation of dsRNA molecule formed by binding of anti-sense RNA molecule to the target mRNA?
(1) RNA interference (2) Gene cloning
(3) Autoradiography (4) Gene flow

180. Read the following statements.
Statement A: The main challenge for producing human insulin by genetic engineering was getting insulin assembled into mature form.
Statement B: Naturally mature insulin is formed after removal of C-chain from proinsulin.
Select the **correct** option.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Both the statements A and B are incorrect

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